

RESEARCH ARTICLE

Integrating multiple energy storage in 1D–2D bridged array carbon-based phase change materials

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Abstract

In response to global energy scarcity, frontiers in multifunctional composite phase change materials (PCMs) with photo-/electro-/magnetothermal triggers show great potential in multiple energy utilization. However, most available composite PCM candidates are inadequate for multiple energy storage applications simultaneously. Herein, a green synthetic route is proposed to develop bimetallic zeolitic imidazolate framework (ZIF)-derived 1D–2D bridged array carbon-based composite PCMs for simultaneous photo-/electro-/magnetothermal energy storage applications. As graphitization-induced catalyst, Co nanoparticles greatly boost the formation of ZIF-derived 1D–2D bridged array carbon framework with high graphitization and low interface electrical/thermal resistance. Benefiting from the broadband intense photon capture, fast photon/electron/phonon transport, and surface plasma resonance effect of Co nanoparticles, the resulting composite PCMs integrate advanced photo-, electro-, and magnetothermal storage functions. Furthermore, composite PCMs also exhibit long-lasting shape stability, structural stability, thermal storage stability, and photo-/electro-/magnetothermal storage stability after undergoing multiple heating–cooling cycles. This study is promising to accelerate the major breakthroughs of advanced multifunctional carbon-based composite PCMs toward multiple energy utilization.

KEYWORDS

electrothermal conversion, magnetothermal conversion, phase change material, photothermal conversion, thermal energy storage

1 | INTRODUCTION

As clean and sustainable energy storage materials, phase change materials (PCMs) are capable of charging or discharging thermal energy through the isothermal phase transition,^{1,2} showing a wide range of applications in dif-

ferent scenarios, such as waste heat recovery,³ device temperature controlling,⁴ building air-conditioning,⁵ smart textile,⁶ drug delivery,⁷ and infrared stealth.⁸ In particular, organic polyethylene glycol (PEG) is advantageous over other types of PCMs due to its high latent heat storage density, adjustable phase change temperature, good thermal stability, and low corrosivity and toxicity.⁹ Nevertheless, the liquid leakage of PCMs above the melting point

Xiao Chen and Jianhang Xu contributed equally to this paper.

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has severely hindered their large-scale applications. Shape stabilization strategy by encapsulating PCMs in porous supporting materials with high specific surface area and large pore volume to prepare shape-stabilized composite PCMs through capillary effect, surface tension, and hydrogen bonding is an effective leakage-proof solution.^{10–12} In this regard, porous supporting materials usually not only play a leakage-proof role in the encapsulation of PCMs, but also endow the composite PCMs with other interesting functions.

Recently, versatile porous supporting materials for PCMs have taken a big step forward in terms of functional customization, mainly including silica,¹³ graphite,^{14,15} carbon nanotubes (CNTs),¹⁶ graphene,¹⁷ boron nitride,¹⁸ metal–organic frameworks,¹⁹ aerogels,²⁰ and MXenes.²¹ Among them, many porous supporting materials themselves enable photo- and electrothermal conversion functions or can also realize the functions of photo-, electro-, and magnetothermal conversion by integrating photosensitizers, thermally conductive agents, electrically conductive agents, and magnetic response agents.^{22–30} Generally, 2D supporting materials are advantageous over 1D and 3D supporting materials for the encapsulation of PCMs, especially graphene and MXenes, due to their more exposed absorption sites, broadband intense absorption capacity, high thermal, and electrical conductivity in-plane.^{31–36} Although graphene- and MXene-based composite PCMs for photo- and electrothermal conversion and storage have made great progress and obtain excellent comprehensive thermophysical properties,^{37–44} high cost and harsh preparation process of graphene and MXenes are not suitable for large-scale application. In addition, the thermal and electrical conductivity is relatively low between the layers of 2D supporting materials due to their large interface thermal and electrical resistance,⁴⁵ which are not conducive to obtaining higher photo- and electrothermal conversion efficiency. It is also worth noting that most available 2D composite PCMs are difficult to integrate photo-/electro-/magnetothermal conversion toward multiple energy utilization. Therefore, the development of advanced multifunctional composite PCMs for simultaneous photo-/electro-/magnetothermal energy conversion and storage still remains a big challenge.

Herein, we proposed a “bridging” strategy to design high thermally and electrically conductive 1D–2D bridged array carbon with interpenetrating network structure via high-temperature calcination of 2D lamellar CoZn-zeolitic imidazolate framework (ZIF). This synthesis strategy of CoZn-ZIF is efficient and green because the synthesis is conducted at room temperature and the solvent used in this experiment is deionized water without any organic solvents, which meets the requirements of green chemistry. During the calcination, Co nanoparticles catalyze the for-

mation of 1D–2D bridged array carbon framework with high graphitization and low interface electrical/thermal resistance, which couples the interaction of photons, electrons, and phonons. These constructed CNT “bridges” among carbon layers guarantee fast photon, electron, and phonon transport in the carbon framework.^{46,47} In addition, Co nanoparticles can also act as a magnetic response unit to trigger the magnetothermal conversion. Therefore, the resulting composite PCMs simultaneously integrate advanced photo-, electro-, and magnetothermal conversion and storage multifunction, as well as long-term thermal durability, showing great potential in multiple energy utilization.

2 | RESULTS AND DISCUSSION

2.1 | Structural analysis of Co/TLC

The preparation schematic of 1D–2D bridged array carbon-based composite PCMs is displayed in Figure 1A. By comparing the morphologies of CoZn-ZIF with different kinds of cobalt and zinc salts, it is found that zinc and cobalt acetates are more suitable for the preparation of 2D lamellar CoZn-ZIF. Notably, the solvent used in this experiment is deionized water, which is a green synthetic route. Specifically, a mixture of 2-methylimidazole solution and cobalt–zinc acetate solution was stirred under room temperature for 2 h to obtain 2D lamellar CoZn-ZIF (Figure 1B and Figures S1 and S2). During the subsequent calcination of CoZn-ZIF, zinc was evaporated, but cobalt was retained. 2D layered stacking structure of CoZn-ZIF is still maintained after high-temperature calcination. During the calcination, cobalt and zinc ions were reduced to simple substances, wherein the elemental zinc was evaporated and discharged with the atmosphere, whereas the elemental cobalt was precipitated from the layered carbon and aggregated into metal nanoparticles to remain in the material. As a graphitization catalyst for carbon at high temperature, cobalt induces the formation of massive CNTs intercalated and penetrated between the layered carbons, forming a network structure of 1D CNTs and 2D carbon layer (Figure 1C–E and Figures S3–S6). The resulting 1D CNTs bridge 2D lamellar carbon to construct fast thermally and electrically conductive pathways with interpenetrating network structure. Meanwhile, transmission electron microscope (TEM) images indicate that the generated cobalt nanoparticles uniformly dispersed inside 2D lamellar carbon (Figure 2A–C and Figures S7 and S8).

The calcined product (cobalt/tube-lamellar-carbon) of CoZn-ZIF was defined as Co/TLC. Compared with CoZn-ZIF, the special array structure of Co/TLC strengthens the encapsulation ability of PCMs and improves the light

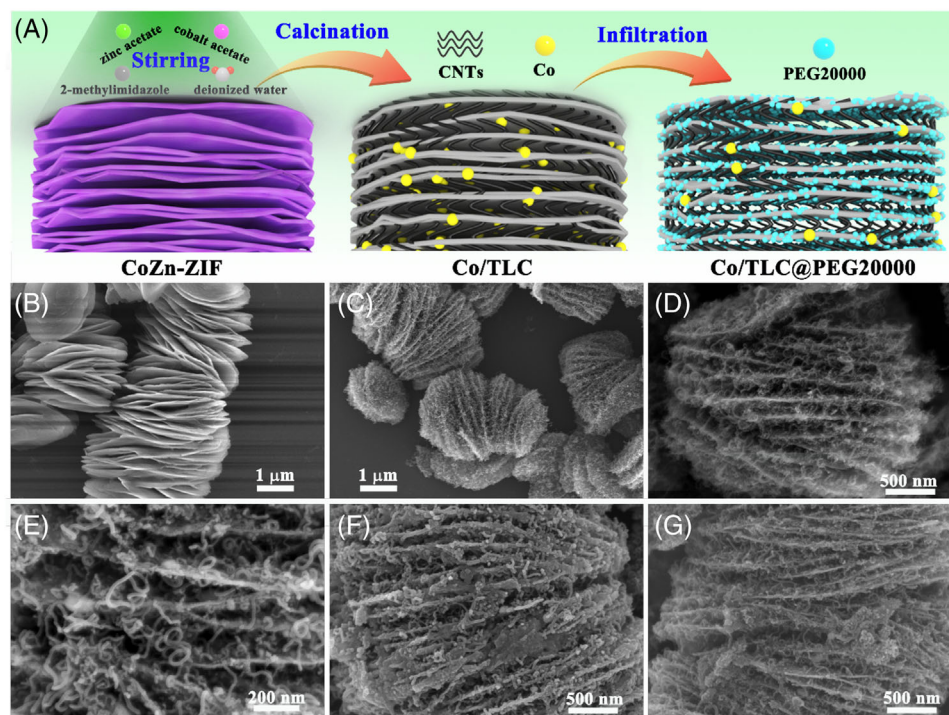


FIGURE 1 Structural analysis: (A) preparation schematic of 1D–2D bridged array carbon-based Co/TLC@PEG20000 composite phase change materials (PCMs); (B) scanning electron microscopy (SEM) image of CoZn-zeolitic imidazolate framework (ZIF); (C–E) SEM images of Co/TLC; (F and G) SEM images of Co/TLC@PEG20000 before and after thermal cycles.

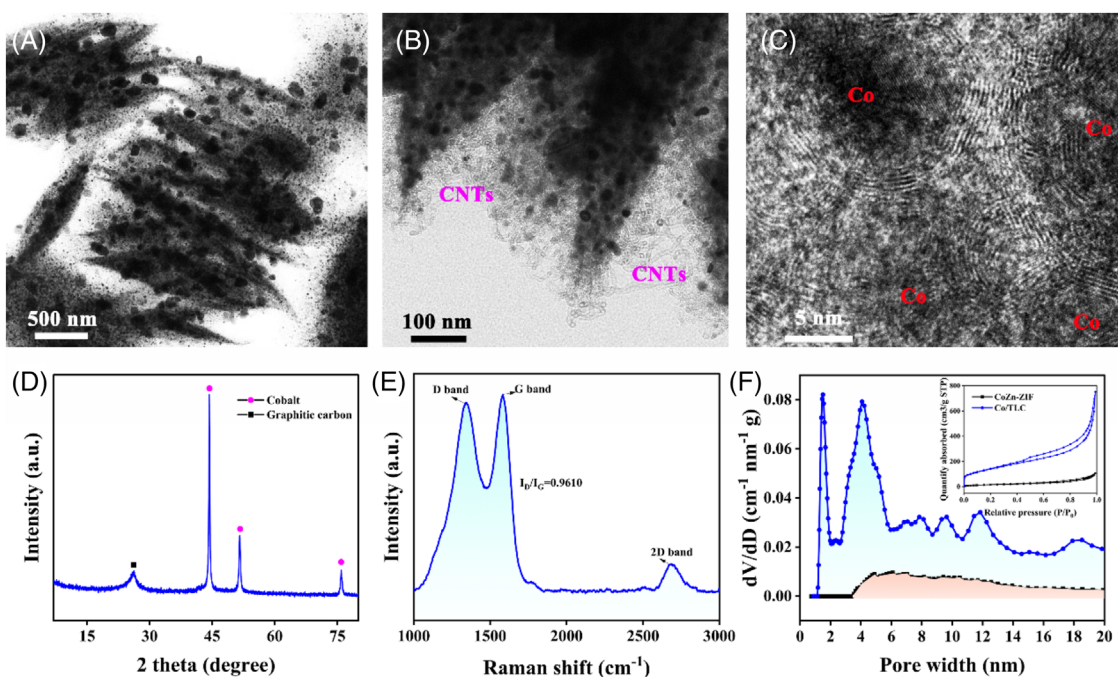


FIGURE 2 Structural analysis: (A–C) transmission electron microscope (TEM) images of Co/TLC; (D) X-ray diffraction (XRD) and (E) Raman spectra of Co/TLC; (F) nitrogen adsorption–desorption isotherms and pore size distribution of Co/TLC and CoZn-zeolitic imidazolate framework (ZIF).

absorption ability. In addition, 1D–2D bridged array carbon network can synergistically improve the out-of-plane and in-plane thermal and electrical conductivity. Cobalt nanoparticles uniformly distributed in TLC can endow the material with good magnetic responsive function. Therefore, after the vacuum infiltration of PEG20000 into 1D–2D bridged Co/TLC, the obtained Co/TLC@PEG20000 composite PCMs (Figure 1F) integrated photo-/electro-/magnetothermal energy conversion and storage functions toward multiple energy utilization.

Phase component analysis of Co/TLC was carried out by X-ray diffraction (XRD) (Figure 2D). Three sharp peaks are located at 44.5° , 51.5° , and 75.8° respectively, which are attributed to the (111), (200), and (220) crystal planes of cobalt, indicating that cobalt ions were successfully reduced. The sharp characteristic peak at 26.0° demonstrates that a large amount of graphitic carbon was formed. Additionally, Raman spectra further reveal the structure of carbon in Co/TLC (Figure 2E). The intensity of D band at 1350 cm^{-1} indicates the number of defects of carbon, whereas that of G band at 1590 cm^{-1} represents the amount of graphitic carbon. The calculated I_D/I_G value is 0.9610, implying that the carbon in Co/TLC reaches a high degree of graphitization. 2D band is the characteristic peak of all graphitic carbons (CNT, graphene, etc.), so the appearance of 2D peak in Co/TLC further demonstrates the highly graphitic carbon structure. Highly graphitic array carbon is more conducive to converting light energy into thermal energy through the coupling mechanism among photons, electrons, and phonons, thus enhancing the photothermal conversion ability.^{48,49}

The pore structure of Co/TLC was measured by nitrogen adsorption and desorption experiment. The calculated pore structure parameters of CoZn-ZIF and Co/TLC are shown in Table S1 and Figure 2F. The isothermal adsorption and desorption curves of CoZn-ZIF and Co/TLC are type IV. CoZn-ZIF is mainly composed of mesopores with a pore volume of $0.16\text{ cm}^3/\text{g}$ and a specific surface area of $58.10\text{ m}^2/\text{g}$. The appearance of H3 hysteresis loop indicates the presence of a large number of stacked mesopores. However, Co/TLC is mainly composed of micropores and mesopores with a pore volume of $1.04\text{ cm}^3/\text{g}$ and a specific surface area of $512.34\text{ m}^2/\text{g}$. Compared with CoZn-ZIF, the specific surface area of Co/TLC is increased by about eight times. The reason is as follows. After the calcination of CoZn-ZIF, a large number of pores are constructed due to the evaporation of zinc. On the other hand, cobalt atoms also escape from their original positions and form pores due to the migration and aggregation of cobalt atoms into nanoparticles. More importantly, CoZn-ZIF-derived numerous CNTs enable the interweaving between CNTs and CNTs, and between CNTs and carbon layers, which greatly enriches the pore structure, improves the porosity

and specific surface area. The constructed 1D–2D bridged Co/TLC can accelerate the fast transmission of phonons and strengthen the shape stability of composite PCMs for leakage-proof (Figure S9).

2.2 | Thermal storage properties

The main characteristic peaks of Co/TLC@PEG20000 composite PCMs are basically consistent with those of Co/TLC and PEG20000, without emerging new characteristic peaks in the composites (Figures S10 and S11). These XRD and Fourier-transform infrared (FTIR) results indicate that only physical interactions occur between Co/TLC and PEG20000 without chemical reaction. In addition, when PEG20000 molecules are physically infiltrated into the pores of Co/TLC, the crystal structures of PEG20000 molecules remain intact, undisturbed by Co/TLC. Compared with Co/TLC, Co/TLC@PEG20000 still maintains original 1D–2D bridged carbon structure. It can be obviously observed that the slit between the carbon layers becomes smooth due to the attachment of PEG20000. Thermophysical parameters (melting temperature, crystallization temperature, melting enthalpy, and crystallization enthalpy) are the basic indexes for evaluating PCMs. Figure 3A shows the differential scanning calorimetry (DSC) curves of PEG20000 and Co/TLC@PEG20000. The DSC curve of PEG20000 has a sharp peak in the heating and cooling stage, corresponding to the heat storage peak and exothermic peak, respectively. Compared with PEG20000, the corresponding heat storage and release peaks of Co/TLC@PEG20000 also emerge at similar temperature nodes, indicating that PEG20000 was successfully infiltrated in Co/TLC. The phase transition process of pure PEG20000 is dominated by homogenous nucleation and growth mechanism. However, Co/TLC can provide numerous heterogeneous nucleation sites for the phase transition of PEG20000. Resultantly, the change in the nucleation pattern results in a slight increase in melting temperature and a slight decrease in the crystallization temperature of Co/TLC@PEG2000 compared with PEG20000. The phase change enthalpy of Co/TLC@PEG20000 is lower than that of pristine PEG20000 due to the occupancy of Co/TLC without latent thermal storage capability. The following formulas are used to calculate the theoretical phase change enthalpy and crystallinity of composite PCMs, respectively:

$$\Delta H_{\text{theory}} = R \times \Delta H_{\text{PEG20000}} \quad (1)$$

$$F_C = \frac{\Delta H_m + \Delta H_f}{\Delta H_{m, \text{theory}} + \Delta H_{f, \text{theory}}} \quad (2)$$

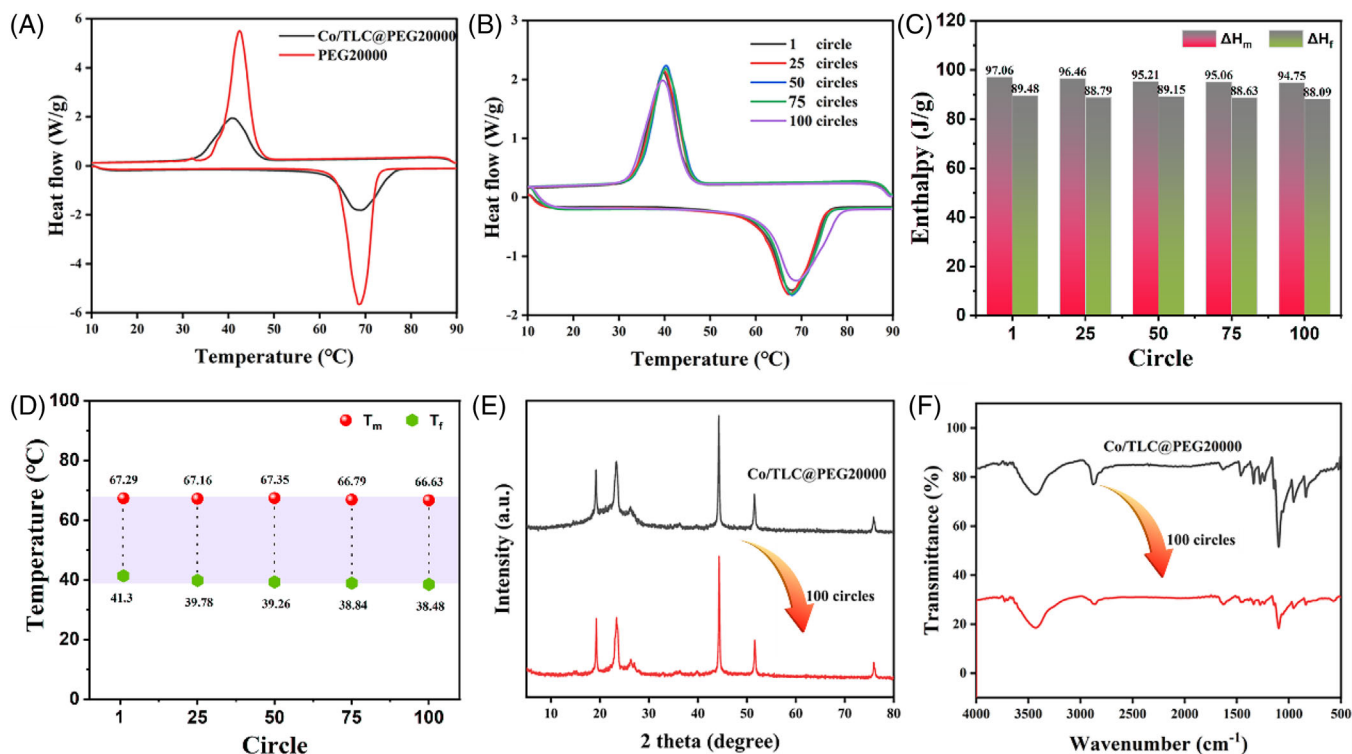


FIGURE 3 Thermal storage properties: (A) differential scanning calorimetry (DSC) curves of PEG20000 and Co/TLC@PEG20000; (B) DSC cycle curves of Co/TLC@PEG20000; (C and D) thermal performance parameters of Co/TLC@PEG20000 after different thermal cycles; (E and F) X-ray diffraction (XRD) and Fourier-transform infrared (FTIR) spectra of Co/TLC@PEG20000 before and after 100 thermal cycles.

where ΔH_{theory} is the theoretical enthalpy of composite PCMs, R is the mass fraction of PEG20000 in the composite PCMs, $\Delta H_{\text{PEG20000}}$ is the phase change enthalpy of pristine PEG20000, F_C is the crystallinity of PEG20000 in the composite PCMs, ΔH_m is the measured melting enthalpy, ΔH_f is the measured solidification enthalpy, $\Delta H_{m, \text{theory}}$ is the theoretical melting enthalpy, and $\Delta H_{f, \text{theory}}$ is the theoretical solidification enthalpy. According to the above formula, the theoretical melting enthalpy and solidification enthalpy of Co/TLC@PEG20000 are 105.93 and 105.30 J/g, respectively, whereas the corresponding measured values by means of DSC are 97.06 and 89.48 J/g, respectively, indicating that PEG20000 molecules do not completely crystallize in Co/TLC. The corresponding crystallinity is calculated to be 88.31%. The incomplete crystallization of Co/TLC@PEG20000 is due to the lack of sufficient free stretching space for PEG20000 molecules in Co/TLC. Resultantly, the free thermal movement of PEG20000 molecules is limited in Co/TLC, thereby reducing the number of effective PEG20000 molecules that can charge and discharge.

Thermal reliability of composite PCMs in repeated use is an important indicator for application. Therefore, we measured the microscopic morphology, phase composition, functional group, and latent heat storage performance of Co/TLC@PEG20000 after 100 heating-cooling cycles.

As seen from Figure 1G, the microscopic morphology of Co/TLC@PEG20000 still retains the 1D–2D bridged carbon structure without noticeable change after 100 heating-cooling cycles very well. Figure 3B shows different DSC cycling curves of Co/TLC@PEG20000. Their DSC curves exhibit similar phase change behavior, indicating that PEG20000 molecules still exist stably in Co/TLC after 100 heating-cooling cycles. The phase change enthalpies ($\Delta H_m = 94.75$ J/g, $\Delta H_f = 88.09$ J/g) of Co/TLC@PEG20000 only decrease by 2.28% and 1.55% after 100 heating-cooling cycles, respectively (Figure 3C). The melting temperature of Co/TLC@PEG20000 remains at about 67°C, and the solidification temperature remains at about 39°C (Figure 3D). The DSC results indicate the superior thermal storage reliability of Co/TLC@PEG20000. Figure 3E,F shows the XRD and FTIR spectra of Co/TLC@PEG20000 after 100 heating-cooling cycles, and the phase compositions and functional groups are highly consistent with pre-cycle results. The scanning electron microscope (SEM), XRD, and FTIR results indicate the superior structural reliability of Co/TLC@PEG20000. In addition, the introduction of Co/TLC slightly increases the thermal stability of PEG20000 (Figure S12). The thermal conductivity of PEG20000 and Co/TLC@PEG20000 is 0.31 and 0.48 W/m K at room temperature, respectively (Figure S13).

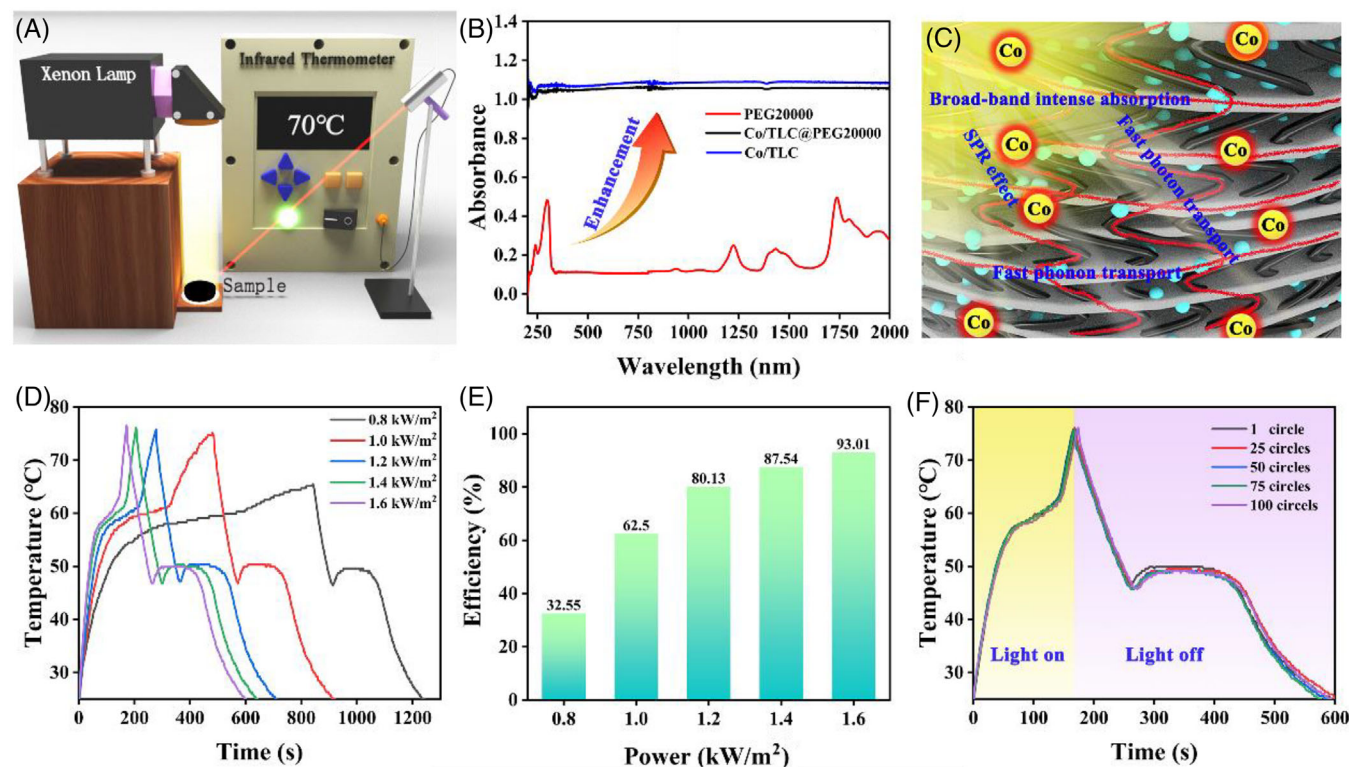


FIGURE 4 Photothermal conversion and storage: (A) optical response test device; (B) UV-Vis-NIR spectra; (C) photothermal conversion mechanism; (D and E) photothermal conversion curves and efficiencies of Co/TLC@PEG20000 under different illumination intensities; and (F) photothermal cycle curves of Co/TLC@PEG20000 after different thermal cycles.

2.3 | Photothermal conversion and storage

The photothermal conversion test method is shown in Figure 4A. The sample is pressed into a disk with a diameter of 19 mm and placed under a xenon lamp with adjustable light intensity. The temperature change curve of the sample is recorded by an infrared thermometer. The absorption spectra of PEG20000 and Co/TLC@PEG20000 were investigated by UV-Vis-NIR spectrophotometer. As shown in Figure 4B, Co/TLC@PEG20000 exhibits high absorption intensity in the whole band. However, PEG20000 exhibits relatively strong absorption in the UV and NIR bands, but the maximum absorption intensity is much lower than that of Co/TLC@PEG20000 in the corresponding band. It is worth noting that PEG20000 absorbs very little visible light, but the proportion of visible light in sunlight is about 50%, meaning that nearly half of the light received by PEG20000 will be reflected under xenon lamp. Therefore, pristine PEG20000 cannot achieve good heating effect in sunlight and is unable to implement photothermal conversion and storage function. The photothermal conversion mechanism of Co/TLC@PEG20000 is shown in Figure 4C, which can be attributed to the special 1D–2D bridged array carbon framework integrating

broadband intense photon capture, fast photon transport, and fast phonon transport, as well as the surface plasma resonance (SPR) mechanism of Co nanoparticles.⁵⁰ As a graphitization-induced catalyst, Co nanoparticles greatly boost the graphitization degree of carbon materials, thus forming high thermally conductive 1D–2D bridged array carbon framework that couples the interaction of photons, electrons and phonons, and associated heat generation.^{48,49} The generated heat can be quickly stored by PEG20000 in the form of latent heat. Meanwhile, the embedded Co nanoparticles in TLC can further boost the light absorption and photothermal conversion due to the SPR effect. Therefore, Co/TLC@PEG20000 exhibits excellent photothermal conversion and storage ability:

$$\eta = \frac{M\Delta H}{PS(t_2 - t_1)} \quad (3)$$

The photothermal conversion efficiency can be calculated according to formula (5–3). η is the photothermal conversion efficiency, M is the mass, ΔH is the latent heat, P is the light intensity, and t_2 and t_1 are the starting and ending temperature points of phase transition that can be obtained by tangent method. Figure 4D shows the photothermal conversion curves of Co/TLC@PEG20000 under

different light intensities (0.8, 1.0, 1.2, 1.4, and 1.6 kW/m²), and the phase transition platform emerges in each curve at 58–63°C in the heating stage and 45–47°C in the cooling stage, indicating that Co/TLC@PEG2000 is capable of carrying out photothermal conversion, storage, and release under light irradiation. It is observed that the temperature of Co/TLC@PEG2000 rises slightly at the cooling platform of each curve because Co/TLC@PEG2000 releases the stored heat energy. On the other hand, the photothermal conversion efficiency of Co/TLC@PEG2000 is higher with the increase of light intensity. Specifically, when the light intensity increases from 0.8 to 1.6 kW/m², the photothermal conversion efficiency increases from 32.55% to 93.01% (Figure 4E) because higher light irradiation shortens the phase transition time and improves the photothermal conversion rate. It is worth mentioning that the photothermal conversion and storage process of Co/TLC@PEG2000 can be triggered by a very small light radiation of 0.8 kW/m². Figure 4F shows the photothermal conversion evolution curves of Co/TLC@PEG2000 during 100 heating–cooling cycles under the light radiation of 1.6 kW/m². The nearly coincident endothermic and exothermic curves indicate the long-term thermal durability of Co/TLC@PEG2000 in solar energy utilization.

2.4 | Electrothermal conversion and storage

In general, organic PCMs are electrical insulators, and their inherent low electrical conductivity prevents them from converting electrical energy into heat through Joule effect. Herein, MOF-derived highly graphitic carbon Co/TLC with good electrical conductivity is compounded with PEG20000 as conductive composite PCMs. As shown in Figure S14, Co/TLC@PEG2000 powder is pressed into a disc with a diameter of 19 mm and then is connected to an LED lamp to form a circuit. After the power is turned on, the LED lamp is lighted up, indicating the flow of current in the circuit containing Co/TLC@PEG2000 disc TLC@PEG20000. This result indicates that our prepared Co/TLC@PEG2000 is a good electrically conductive material, which can trigger the electrothermal conversion and storage through Joule effect. The electrical conductivity of PEG20000 and Co/TLC@PEG2000 is 10^{−13} and 1.2 S/m at room temperature, respectively (Figure S15).

The electrothermal conversion test method is shown in Figure 5A. The disc sample with a diameter of 19 mm is connected to the electrochemical workstation with the assistance of two copper sheets. The thermocouple temperature probe is placed under the copper sheets. The test device is sealed up with foam to prevent the generated heat from being exchanged convectively with the

environment. The electrothermal conversion mechanism of Co/TLC@PEG2000 is shown in Figure 5B, which can be attributed to the special 1D–2D bridged array carbon framework integrating fast electron transport and fast phonon transport. As a graphitization-induced catalyst, Co nanoparticles greatly induce the formation of 1D–2D bridged array carbon framework with high graphitization and low interface electrical/thermal resistance, which couples the interaction of electrons and phonons, and associated heat generation. As an excellent electron and phonon transport medium, the bridged CNTs accelerate the rapid transport of electrons and phonons between the carbon layers and strengthen the overall electrical and thermal conductivity of Co/TLC@PEG2000. The generated heat can be quickly stored by PEG20000 in the form of latent heat. Therefore, Co/TLC@PEG2000 exhibits excellent electrothermal conversion and storage ability.

Figure 5C shows the electrothermal conversion curves of Co/TLC@PEG2000 under different voltages (1.5, 1.7, 1.9, 2.1, and 2.3 V). It can be observed that when the applied voltages are 1.5 and 1.7 V, the temperature of Co/TLC@PEG2000 rises slowly and even is unable to reach the phase transition point. When the voltage rises to a higher value (1.9, 2.1, and 2.3 V), the heating rate of Co/TLC@PEG2000 increases significantly and reaches the phase transition point in a short time. The phase transition heat storage platform appears at 45–50°C, indicating that the electric energy is converted into the thermal energy and stored in PEG20000 in the form of latent heat. When the voltage is removed, the phase transition heat release platform appears at 42–43°C, and the stored latent heat is released into the environment. In addition, the evolution trend of current–time curve (Figure 5D) is consistent with the temperature–time curve (Figure 5C). Specifically, when the voltages are 1.5 and 1.7 V, the current shows a relatively stable state with time, and it is roughly a horizontal line. When the voltages are 1.9, 2.1, and 2.3 V, the current shows an obvious increasing trend, and the current increases faster with the increase of voltage. The explanation is as follows. Carbon material is a negative temperature coefficient thermistor. When the temperature rises, the electrical resistance decreases, and even a small temperature change might cause a large change in the electrical resistance. When the voltages are 1.5 and 1.7 V, the temperature of Co/TLC@PEG2000 rises slowly, and the temperature is low and the current is stable. Therefore, the electrical resistance of Co/TLC@PEG2000 changes little in the whole heating process. When the voltages are 1.9, 2.1, and 2.3 V, the temperature of Co/TLC@PEG2000 rises quickly, and the temperature is high and the current increases rapidly and continuously. Therefore, the electrical resistance of Co/TLC@PEG2000 decreases rapidly. In addition, with the extension of the

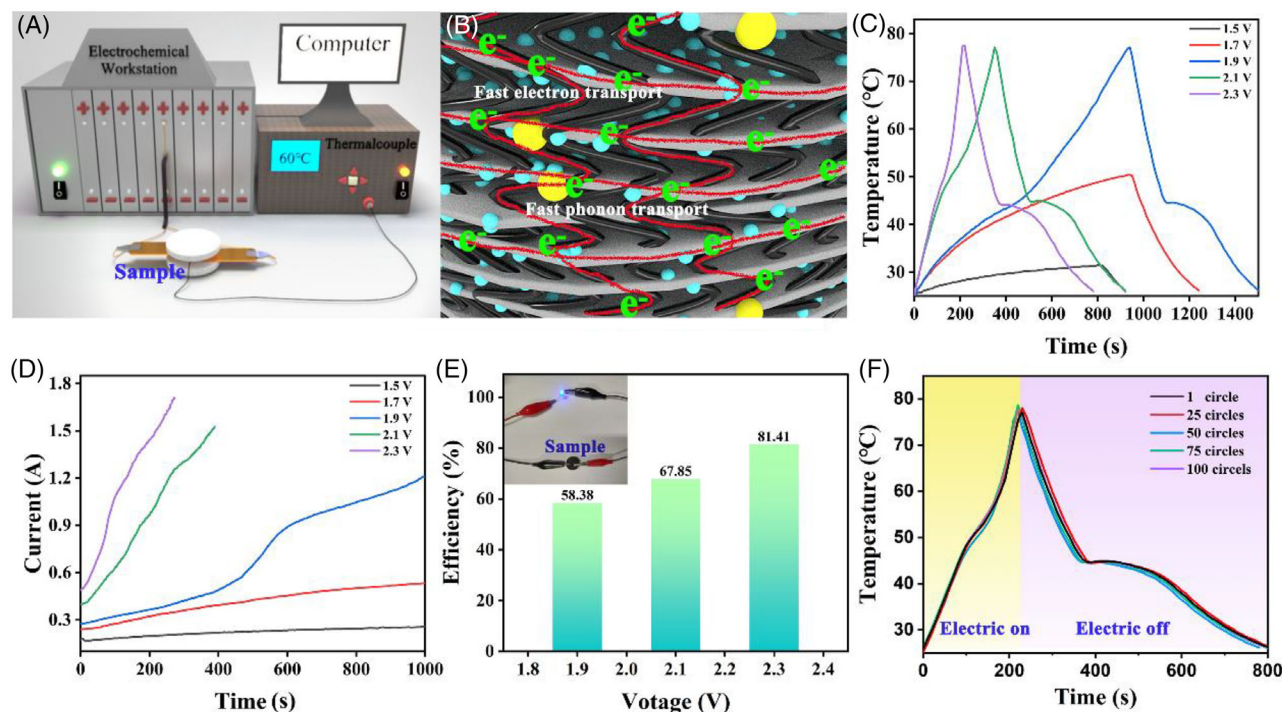


FIGURE 5 Electrothermal conversion and storage: (A) electric response test device; (B) electrothermal conversion mechanism; (C) electrothermal conversion curves of Co/TLC@PEG20000 under different voltages; (D) current variation of Co/TLC@PEG20000 in the loop during electrothermal conversion; (E) electrothermal conversion efficiencies of Co/TLC@PEG20000 under different voltages; and (F) electrothermal cycle curves of Co/TLC@PEG20000 after different thermal cycles.

electrothermal conversion time, the melted PEG20000 can make Co/TLC nanosheets contact more tightly, thereby reducing the electrical resistance and accelerating the electrical conduction. By comparing the current–time curves and temperature–time curves of Co/TLC@PEG20000 at 1.9, 2.1, and 2.3 V, it can be observed that when the solid–liquid phase transition emerges, the corresponding temperature and current curves both exhibit inflection points, indicating that the molecular thermal motion of PEG20000 during the solid–liquid transition influences the electrical resistance of Co/TLC@PEG20000:

$$\eta = \frac{m\Delta H}{W} \quad (4)$$

$$W = qt = \int_{t_1}^{t_2} q(t) dt \quad (5)$$

$$q(t) = UI(t) \quad (6)$$

$$\eta = \frac{m\Delta H}{U \int_{t_1}^{t_2} I(t) dt} \quad (7)$$

The electrothermal conversion efficiency can be calculated according to the formula. η is the electrothermal conversion efficiency, m is the mass, ΔH is the latent heat, W is the output work, q is the output power, U is the input

voltage, I is the output current, and t_2 and t_1 are the starting and ending temperature points of phase transition that can be obtained by tangent method. Figure 5E shows the electrothermal conversion efficiencies of Co/TLC@PEG20000 under different voltages of 1.9, 2.1, and 2.3 V. When the voltages rise from 1.9 to 2.3 V, the electrothermal conversion efficiency rises from 58.38% to 81.41%. This is mainly because the increased input voltage accelerates the heating rate, thus reducing the proportion of heat environmental loss during the electrothermal conversion process and improving the electrothermal conversion efficiency. It is worth mentioning that a small voltage of 1.9 V can trigger the electrothermal conversion and storage of Co/TLC@PEG2000. Figure 5F shows the electrothermal conversion evolution curves of Co/TLC@PEG2000 during 100 heating–cooling cycles under the voltage of 2.3 V. The nearly coincident endothermic and exothermic curves indicate the long-term thermal durability of Co/TLC@PEG2000 in electric energy utilization.

2.5 | Magnetothermal conversion and storage

Iron, cobalt, and nickel metals are common magnetic materials. Under the action of alternating magnetic

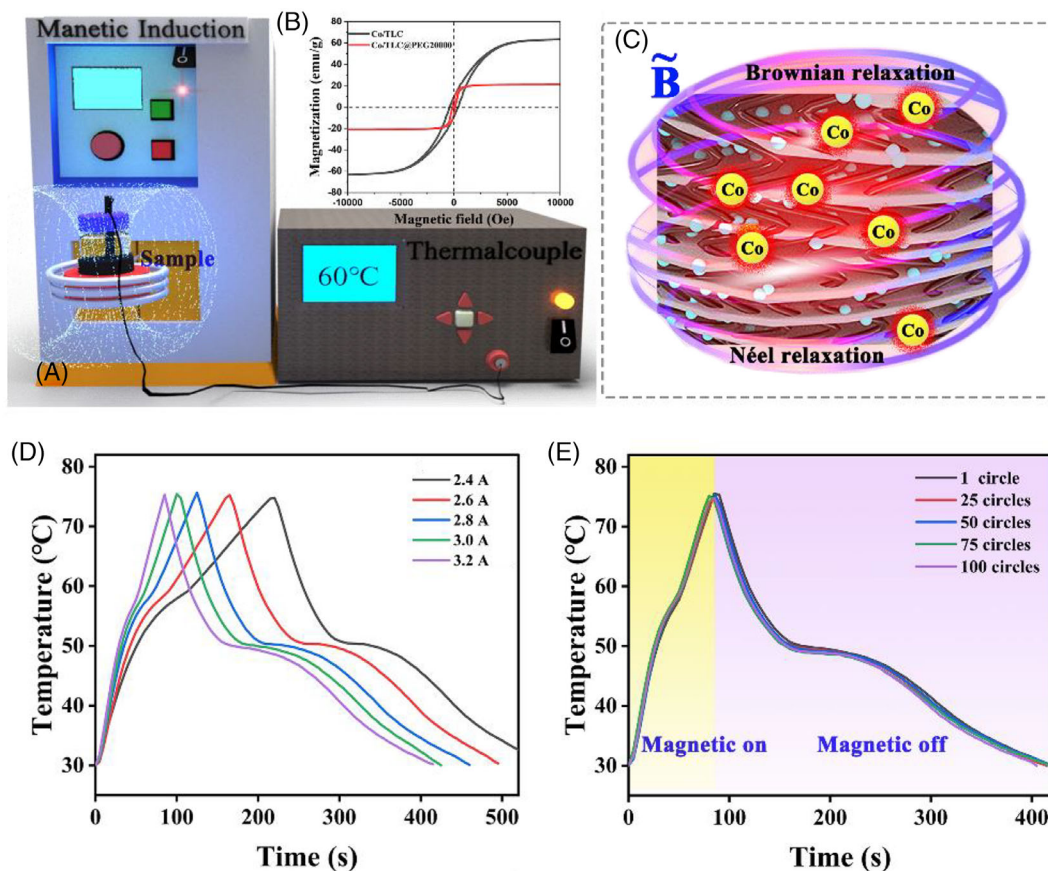


FIGURE 6 Magnetothermal conversion and storage: (A) magnetic response test device; (B) hysteresis loops of Co/TLC and Co/TLC@PEG20000; (C) magnetothermal conversion mechanism; (D) magnetothermal conversion curves of Co/TLC@PEG20000 under different alternating currents; and (E) magnetothermal cycle curves of Co/TLC@PEG20000 under an alternating current of 3.2 A after different thermal cycles.

field, vortex current is generated in the magnetic metal. Therefore, the metal cobalt nanoparticles in Co/TLC@PEG20000 can be used as a nano-heater to convert magnetic energy into thermal energy under the action of alternating magnetic field and stored by PEG20000 in the form of latent heat. The magnetothermal conversion test method is shown in Figure 6A. Figure S16 proves the magnetic property of Co/TLC@PEG20000. Specifically, Co/TLC@PEG20000 is dissolved in the absolute ethanol, and the strip magnet is put aside. Resultantly, Co/TLC@PEG20000 in the solution moves to one side of the magnet, indicating its good magnetism. Further, Figure 6B shows the field-dependent magnetization curves of Co/TLC and Co/TLC@PEG20000 between ± 10 kOe. Ferromagnetic Co/TLC@PEG20000 exhibits a low magnetic retentivity and coercivity. The saturation magnetization of Co/TLC@PEG20000 is 21.1 emu/g. Therefore, Co/TLC@PEG20000 can convert electromagnetic energy into thermal energy through the hysteresis effect in an alternating magnetic field. The magnetothermal conversion mechanism of Co/TLC@PEG20000 is

shown in Figure 6C, which can be attributed to the Néel and Brownian relaxation effects of Co nanoparticles.^{23,32} These Co nanoparticles dispersed in TLC not only act as a magnetic response unit but also act as a catalyst to induce the formation of numerous CNTs that bridge carbon layer. This constructed 1D–2D bridged array carbon framework can accelerate the phonon transport due to the low interface thermal resistance. Therefore, the generated heat through magnetic excitation can be transferred to PEG20000 quickly and stored in the form of latent heat.

The magnetothermal conversion curves of PEG20000 and Co/TLC@PEG20000 are shown in Figure 6D. Under the alternating magnetic field, the temperature of Co/TLC@PEG20000 rises rapidly and reaches the phase transition point, whereas PEG20000 does not respond to the magnetic field. To systematically study the magnetothermal conversion performance of Co/TLC@PEG20000, we measured its magnetothermal conversion curves under different alternating currents (2.4, 2.6, 2.8, 3.0, and 3.2 A). As seen from Figure 6D, the phase change endothermic platform appears in each

temperature curve at 57–60°C, indicating the latent heat storage of PEG20000. After removing the magnetic field, the sample is cooled naturally, and the phase change exothermic platform appears in each temperature curve at 48–50°C, indicating the latent heat release of PEG20000. With the increase of current intensity, the magnetic heating rate increases, resulting in the reduction of phase transition time. In addition, the magnetothermal conversion of Co/TLC@PEG20000 was further tested for 100 heating–cooling cycles under 3.2 A alternating current magnetic field, and each temperature–time curve is highly consistent, indicating its excellent thermal durable stability under the alternating magnetic field (Figure 6E). Therefore, Co/TLC@PEG20000 shows good application prospect in the field of magnetothermal conversion and storage.

3 | CONCLUSION

In this study, 2D lamellar CoZn-ZIF was prepared by an efficient and green method without any organic solvents at room temperature. After high-temperature calcination of CoZn-ZIF, high thermally and electrically conductive 1D–2D bridged array carbon with interpenetrating network structure was constructed under the catalytic action of Co nanoparticles. The 1D–2D bridged array carbon integrates broadband intense photon capture, fast photon transport, fast electron transport, and fast phonon transport due to high graphitization, intense light capture, and low interface electrical/thermal resistance. Therefore, Co/TLC@PEG20000 exhibits excellent photothermal (93.01%) and electrothermal (81.41%) conversion and storage ability. It is worth noting that the generated Co nanoparticles uniformly dispersed inside 2D lamellar carbon can further boost the light absorption and photothermal conversion due to the SPR effect. In addition, these Co nanoparticles can act as a magnetic response unit to trigger the magnetothermal conversion and storage function of Co/TLC@PEG20000 through the Néel and Brownian relaxation effects. Furthermore, Co/TLC@PEG20000 exhibits excellent shape stability, structural stability, thermal storage stability, and photo-/electro-/magnetothermal conversion stability after 100 heating–cooling cycles, showing great potential in multiple energy utilization.

4 | EXPERIMENTAL SECTION

4.1 | Materials

Cobalt and zinc acetates were purchased from Guangdong Guanghua Sci-Tech Co., Ltd. PEG (PEG20000) and

2-methylimidazole were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Ethanol was purchased from Sinopharm Chemical Reagent Beijing Co. Ltd.

4.2 | Synthesis of CoZn-ZIF

First, 3.936 g 2-methylimidazole was dissolved in 100 mL deionized water to obtain solution A. An amount of 1 g cobalt and 0.5 g zinc acetates were dissolved in 10 mL deionized water to obtain solution B. Then, solution B was poured into solution A, and the corresponding mixture was kept stirring at room temperature for 2 h. Finally, the product was washed with water several times and freeze-dried for 24 h to obtain CoZn-ZIF powder.

4.3 | Synthesis of Co/TLC

The alumina boat containing CoZn-ZIF powder was placed in the tubular furnace and heated up to 900°C in hydrogen argon mixture. After the furnace was cooled to room temperature, the obtained product is defined as Co/TLC (cobalt/tube-lamellar-carbon).

4.4 | Synthesis of Co/TLC@PEG20000

Co/TLC@PEG20000 was prepared by solution impregnation. First, Co/TLC and PEG20000 were mixed in absolute ethanol at 80°C. Then, the mixture was sonicated for 30 min to ensure that PEG20000 was completely dissolved in absolute ethanol and homogeneously mixed with Co/TLC powder. Subsequently, the solution was kept in the oven at 80°C for 12 h to remove ethanol. The obtained composite is defined as Co/TLC@PEG20000. In this experiment, the mass ratio of Co/TLC to PEG20000 is set to 4:6.

4.5 | Characterizations

The morphology of samples was observed using an SEM (Regulus 8100) and TEM (JEM-2200FS). The crystal structure of samples was characterized with XRD (Bruker D8-Advance, Cu K_{α} radiation). The graphitization degree was investigated with Raman (Renishaw inVia) under the excitation wavelength of 532 nm. N_2 adsorption and desorption isotherm was collected using a Micromeritics ASAP2460 instrument. The FTIR spectra were recorded on a Nicolet 6700. The adsorption spectra of samples were recorded by an ultraviolet visible spectrophotometer. The heat storage capacity was characterized with DSC (DSC3, Mettler

Toledo) at a heating and cooling rate of 10°C/min. The photothermal conversion performance was investigated by a solar simulator (xenon lamp). The magnetic properties were evaluated at room temperature with a vibrating sample magnetometer (VSM, Quantum Design). A magnetic induction heating equipment was used to obtain the temperature–time curves of samples under alternating magnetic fields. The electrothermal conversion performance was investigated by a Princeton electrochemical workstation (PMC 1000 and 500).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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